

AREA BETWEEN TWO CURVES

Suppose $f(x) \geq g(x)$ for $a \leq x \leq b$

Approximate area of k th subinterval
where x_k is a point.

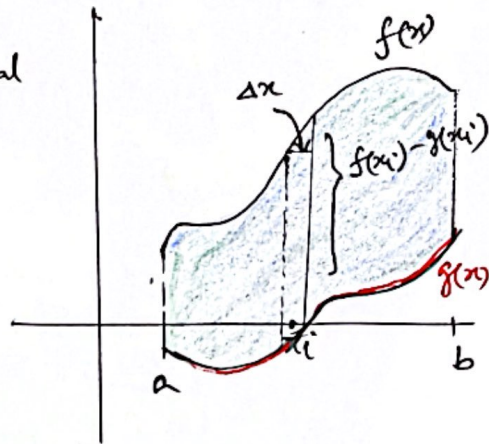
$$A_k \approx [f(x_k) - g(x_k)] \Delta x_k$$

So, the Riemann sum that
approximates the Area,

$$A \approx \sum_{k=1}^n [f(x_k) - g(x_k)] \Delta x_k$$

Now, the exact area $A = \lim_{\Delta x \rightarrow 0} \sum_{k=1}^n [f(x_k) - g(x_k)] \Delta x_k$

$$= \int_a^b [f(x) - g(x)] dx$$



— x —

PROBLEM: Find the area of the region that is enclosed between the curves $y = x^2$ and $y = x + 6$.

Solⁿ: Here $y = x^2$ ——— (i)
 $y = x + 6$ ——— (ii)

solving (i) & (ii), we get

$$x^2 = x + 6 \Rightarrow (x+2)(x-3) = 0$$

$$x = -2, x = 3$$

$\therefore$ the intersection points are $(-2, 4)$ & $(3, 9)$.

For x -axis, upper curve is $y = x + 6$ and lower curve is $y = x^2$ and area bounded by -2 to 3 .

$$\text{Now, Area} = \int_{-2}^3 [(x+6) - x^2] dx$$

$$= \left(\frac{x^2}{2} + 6x - \frac{x^3}{3} \right) \Big|_{-2}^3$$

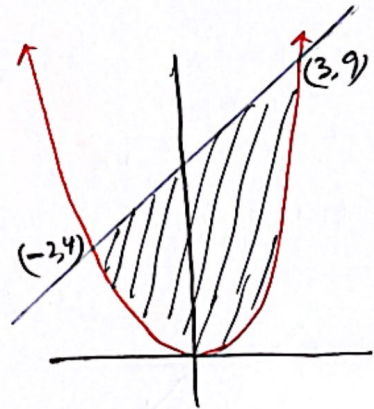
$$= \left(\frac{9}{2} + 18 - \frac{27}{3} \right) - \left(\frac{4}{2} - 12 + \frac{8}{3} \right)$$

$$= \frac{27}{2} + \frac{22}{3} = \frac{125}{6}$$

✱

PROBLEM: Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

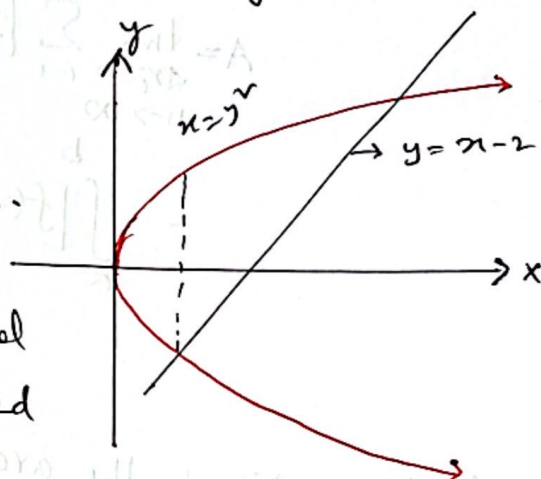
PROBLEM: Find the area of the region in the first quadrant that is bounded by above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$.



$$\begin{aligned}
 \text{Now, } A &= \int_{-2}^3 [(x+6) - x^2] dx = \left(\frac{x^2}{2} + 6x - \frac{x^3}{3} \right) \Big|_{-2}^3 \\
 &= \left(\frac{3^2}{2} + 6 \cdot 3 - \frac{3^3}{3} \right) - \left[\frac{(-2)^2}{2} + 6(-2) - \frac{(-2)^3}{3} \right] \\
 &= \left(\frac{9}{2} + 18 - \frac{27}{3} \right) - \left(2 - 12 + \frac{8}{3} \right) \\
 &= \frac{81}{6} + \frac{24}{3} = \frac{125}{6} \quad *
 \end{aligned}$$

Problem: Find the area of the region enclosed by $x = y^2$ and $y = x - 2$.

Solⁿ: Here we can solve it ^{by} two ways
 by ~~not~~ based on x-axis and y-axis.
 We need to know which function
 we want to say upper curve and
 which one is lower curve based
 on axis.



the solⁿ is in the book.

page 417	H. Anton
Example 4	

For y-axis

$$x = y^2, x = y + 2$$

$$\text{so, } y^2 - y - 2 = 0 \Rightarrow (y+1)(y-2) = 0, y = -1, 2$$

so, intersection points are $(1, -1), (4, 2)$

$$\begin{aligned}
 A &= \int_{-1}^2 [y+2 - y^2] dy = \left(\frac{y^2}{2} + 2y - \frac{y^3}{3} \right) \Big|_{-1}^2 = \left(\frac{2^2}{2} + 2 \cdot 2 - \frac{2^3}{3} \right) - \left[\frac{(-1)^2}{2} + 2(-1) - \frac{(-1)^3}{3} \right] \\
 &= \frac{9}{2} \quad *
 \end{aligned}$$

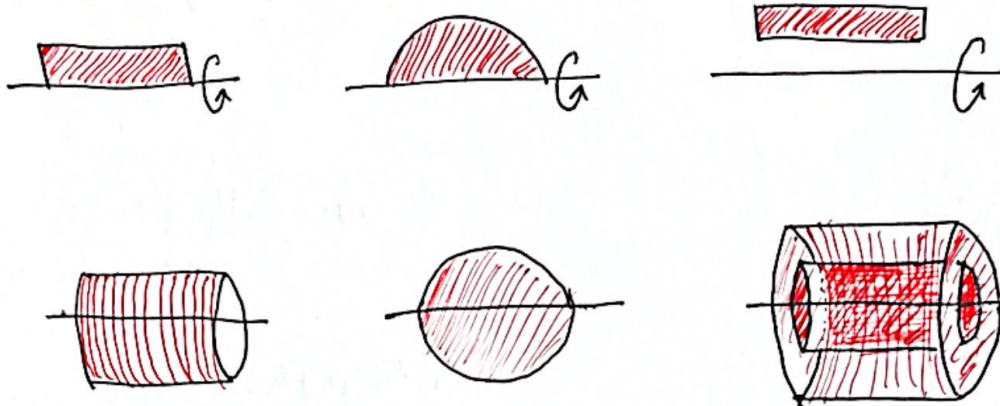
EXTRA PROBLEMS:

~~Problem~~ Book: H. Anton

page: 419

Problems: 7-26

Solids of Revolution



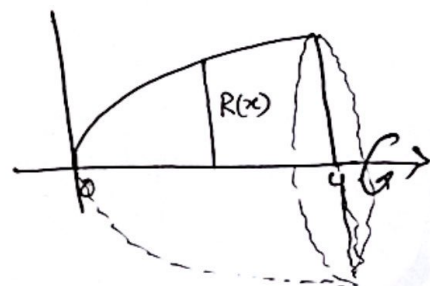
Volume by disks for Rotation About the x-axis

$$V = \int_a^b A(x) dx = \int_a^b \pi [R(x)]^2 dx \quad \text{here } R(x) = \text{Radius of a cross-section circular disk.}$$

PROBLEMS: The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$, and the x-axis is revolved about the x-axis to generate a solid. Find its volume.

Solⁿ:

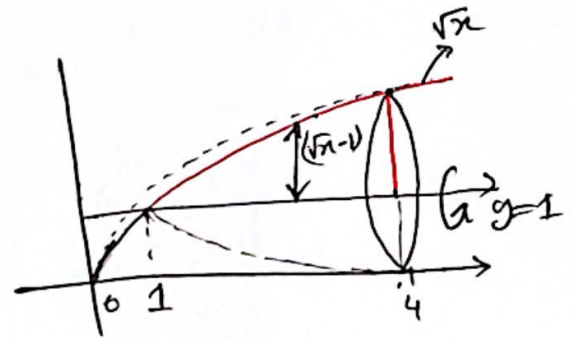
$$\begin{aligned} V &= \int_0^4 \pi [R(x)]^2 dx \\ &= \int_0^4 \pi (\sqrt{x})^2 dx \\ &= \pi \left. \frac{x^2}{2} \right|_0^4 = \pi \cdot \frac{4^2}{2} = 8\pi \end{aligned}$$



PROBLEMS: Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1$, $x = 4$ about the line $y = 1$.

Sol.

$$\begin{aligned}
 V &= \int_1^4 \pi [R(x)]^2 dx \\
 &= \int_1^4 \pi (\sqrt{x} - 1)^2 dx \\
 &= \pi \int_1^4 (x - 2\sqrt{x} + 1) dx \\
 &= \pi \left(\frac{x^2}{2} - 2 \cdot \frac{2}{3} x^{3/2} + x \right) \Big|_1^4 \\
 &= \frac{7\pi}{6}
 \end{aligned}$$



$y = \sqrt{x}$, $y = 1$
 $\sqrt{x} = 1$, $x = 4$
 $x = 1$
 So, about x -axis i.e.
 for dx ; the
 domain is $[1, 4]$.

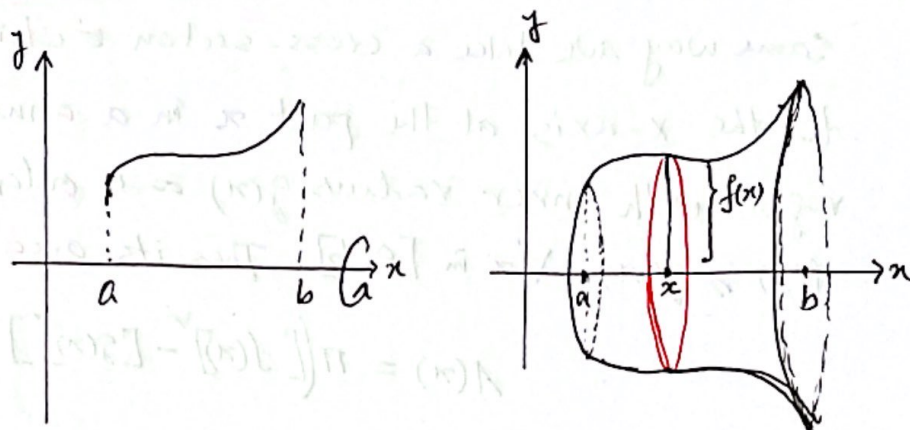
EXTRA PROBLEMS

H. ANTON

page: 429

PROBLEMS: 11-18, 21-26

VOLUMES BY DISKS PERPENDICULAR TO THE x-AXIS



Taking a cross section which is perpendicular to the x-axis at a point x is a circular disk of radius $f(x)$. Then the area of this region is

$$A(x) = \pi [f(x)]^2$$

So, the volume of the solid is

$$V = \int_a^b \pi [f(x)]^2 dx$$

EXAMPLE: Find the volume of the solid that is obtained when the region under the curve $y = \sqrt{x}$ over the interval $[1, 4]$ is revolved about the x-axis.

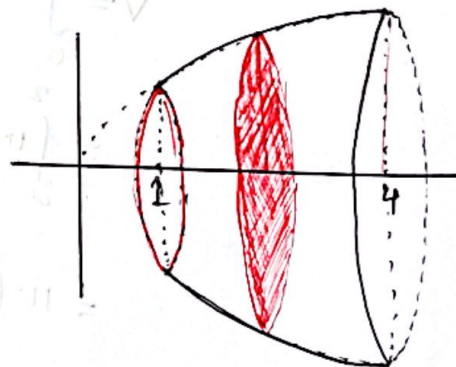
Solⁿ

Volume,

$$V = \int_1^4 \pi [f(x)]^2 dx$$

$$= \int_1^4 \pi [\sqrt{x}]^2 dx = \int_1^4 \pi x dx$$

$$= \pi \left[\frac{x^2}{2} \right]_1^4 = 18\pi - \frac{\pi}{2} = \frac{35\pi}{2}$$



✗

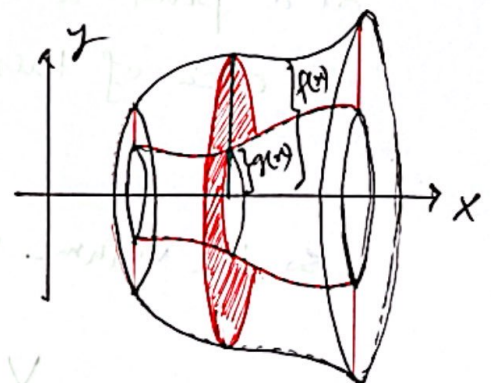
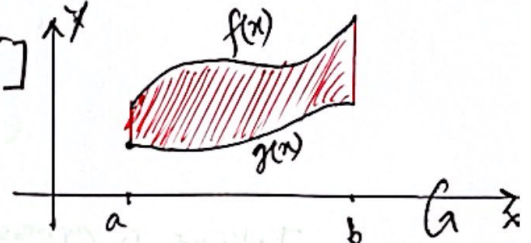
VOLUMES BY WASHERS PERPENDICULAR TO THE X-AXIS

Same way, we take a cross-section which is perpendicular to the x-axis at the point x in an annular or "washer-shaped" region with inner radius $g(x)$ and outer radius $f(x)$ where $f(x) \geq g(x) \quad \forall x \in [a, b]$. Then its area is

$$A(x) = \pi([f(x)]^2 - [g(x)]^2)$$

The volume of the solid is

$$V = \int_a^b \pi([f(x)]^2 - [g(x)]^2) dx$$



PROBLEM: Find the volume of the solid generated when the region between the eq's $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ over the interval $[0, 2]$ is revolved about the x-axis.

Solⁿ: Volume

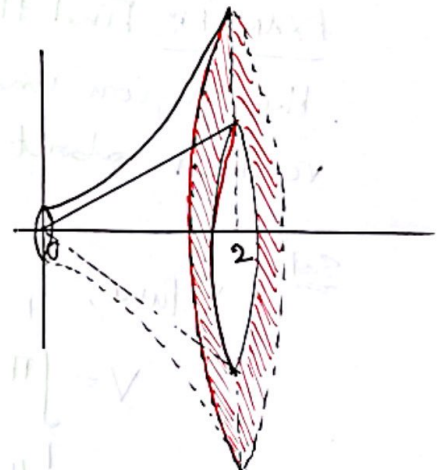
$$V = \int_0^2 \pi \left[\left(\frac{1}{2} + x^2 \right)^2 - x^2 \right] dx$$

$$= \int_0^2 \pi \left(\frac{1}{4} + x^4 \right) dx$$

$$= \pi \left(\frac{x}{4} + \frac{x^5}{5} \right) \Big|_0^2$$

$$= \frac{69\pi}{10}$$

✱

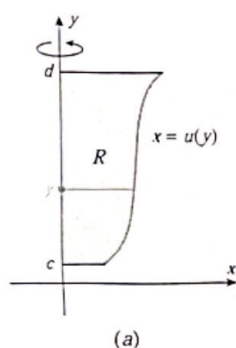


EXTRA PROBLEMS :

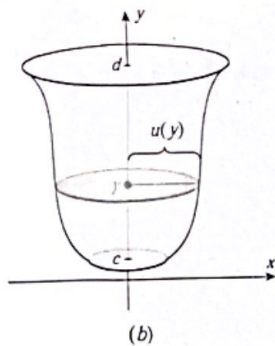
REFERENCE BOOK: H. ANTON

PAGE : 429

PROBLEMS: 11-18 , 21-26 , 33-34, 39, 40, 42, 43, 44



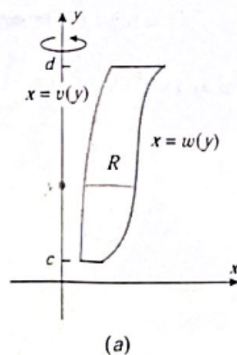
(a)



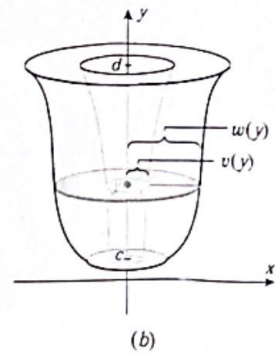
(b)

Disks

▲ Figure 6.2.14



(a)



(b)

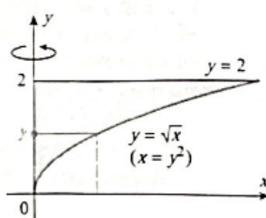
Washers

▲ Figure 6.2.15

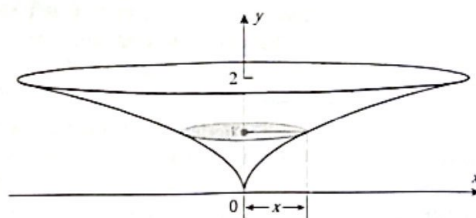
► **Example 5** Find the volume of the solid generated when the region enclosed by $y = \sqrt{x}$, $y = 2$, and $x = 0$ is revolved about the y -axis.

Solution. First sketch the region and the solid (Figure 6.2.16). The cross sections taken perpendicular to the y -axis are disks, so we will apply (7). But first we must rewrite $y = \sqrt{x}$ as $x = y^2$. Thus, from (7) with $u(y) = y^2$, the volume is

$$V = \int_c^d \pi[u(y)]^2 dy = \int_0^2 \pi y^4 dy = \frac{\pi y^5}{5} \Big|_0^2 = \frac{32\pi}{5} \quad \blacktriangleleft$$



► Figure 6.2.16



OTHER AXES OF REVOLUTION

It is possible to use the method of disks and the method of washers to find the volume of a solid of revolution whose axis of revolution is a line other than one of the coordinate axes. Instead of developing a new formula for each situation, we will appeal to Formulas (3) and (4) and integrate an appropriate cross-sectional area to find the volume.

► **Example 6** Find the volume of the solid generated when the region under the curve $y = x^2$ over the interval $[0, 2]$ is rotated about the line $y = -1$.

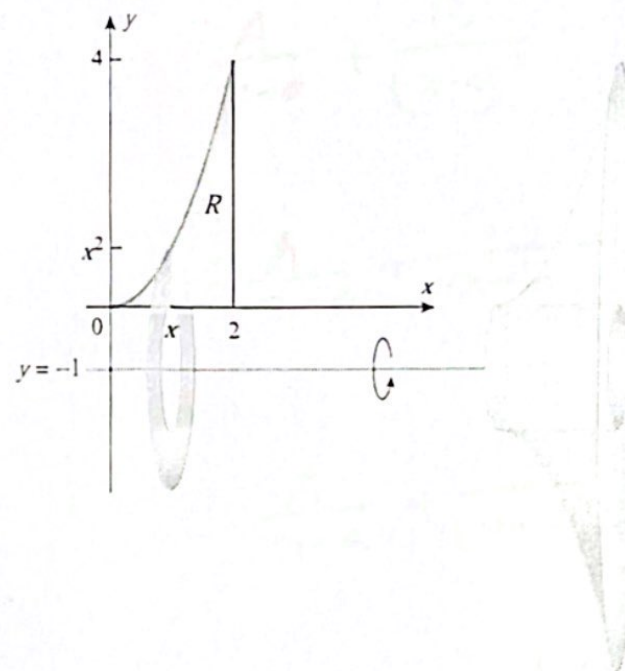
Solution. First sketch the region and the axis of revolution; then imagine revolving the region about the axis (Figure 6.2.17). At each x in the interval $0 \leq x \leq 2$, the cross section of the solid perpendicular to the axis $y = -1$ is a washer with outer radius $x^2 + 1$ and inner radius 1. Since the area of this washer is

$$A(x) = \pi[(x^2 + 1)^2 - 1^2] = \pi(x^4 + 2x^2)$$

428 Chapter 6 / Applications of the Definite Integral in Geometry, Science, and Engineering

it follows by (3) that the volume of the solid is

$$V = \int_0^2 A(x) dx = \int_0^2 \pi (x^4 + 2x^2) dx = \pi \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 \right]_0^2 = \frac{176\pi}{15} \quad \blacktriangleleft$$



► Figure 6.2.17